

Unit 13, Lesson 8: Creating Displays for Numerical Data

Benchmark

MA.7.DP.1.5 Given a real-world numerical or categorical data set, choose and create an appropriate graphical representation.

Learning Target

- ☐ I can create an appropriate graphical representation for a numerical data set.

13.8.1 Warm-Up: Mystery Boxes

1. The data set below contains 14 numbers, listed in order from least to greatest. Six of the values have been removed. Your job is to use the clues given to determine the missing values. Write your answers in the boxes.

- The mean is 26.
- The median is 22.
- The interquartile range is 20.
- The mode is 18.
- The range is 60.

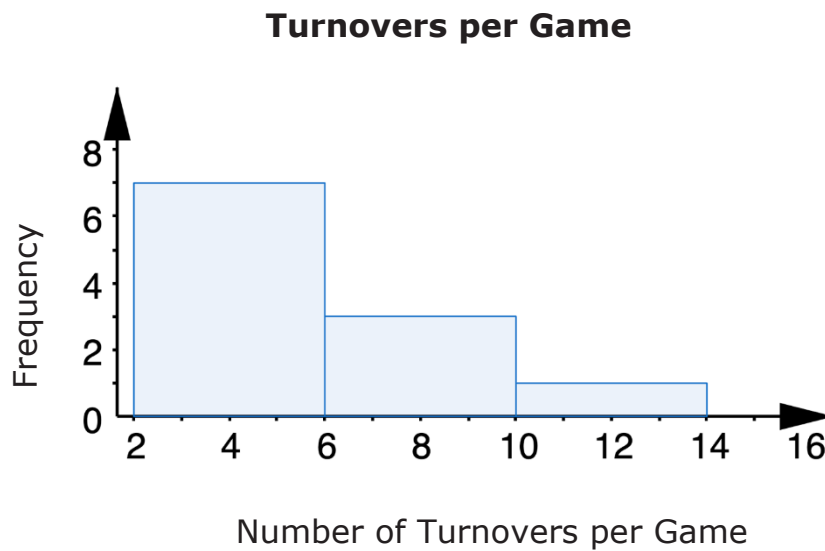
1		4		15	18			29	30	32		58	
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13.8.2 Collaboration: Who Made it Best?

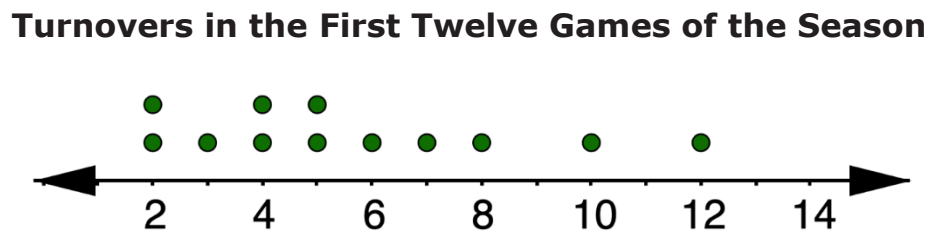
1. Dante and Lorraine are both managers of the seventh-grade basketball team. They are trying to decide which graphical representation to use to show the number of turnovers the team has had per game in the first 12 games of the season, shown in the data set below.

$\{2, 2, 3, 4, 4, 5, 5, 6, 7, 8, 10, 12\}$

Lorraine made a histogram, as shown.



Dante made a line plot, as shown.



- a. Which of the two graphical representations shown is best to display the data?
- b. Explain your reasoning.
- c. Compare your work with your partner and if necessary, resolve any differences.
- d. Discuss with your partner whether a box plot or stem-and-leaf plot would be useful for displaying this data set. Summarize your discussion.

13.8.3 Try It: Creating Graphical Representations

1. Laverne and Rocco planned to travel the greatest distance between any two points in the United States of America — from Log Point, Elliot Key, Florida to Kure Atoll, Hawaii — a distance of 5,859 miles.

They took a ferry from Log Point to Miami, then drove to Los Angeles, California, where they took a plane to Hawaii. They drove anywhere from 5-8 hours a day, and kept track of how many miles they traveled per day on their trip. This information is shown below.

$\{224, 294, 309, 317, 321, 339, 355, 361, 383, 456, 2500\}$

- a. Create an appropriate graphical representation for the data.
 - b. If the value 2500 were removed from the data set, which display would you create to represent the data? Explain your reasoning.
 - c. Create an appropriate graphical representation for the data without the value 2500.
2. An earthworm farmer set up several containers of a certain species of earthworm so he could learn about their lengths. Lengths of earthworms provide information about their ages. The farmer measured the lengths of the earthworms in one container.

The farmer recorded the earthworms' lengths, in millimeters (mm), in the table shown.

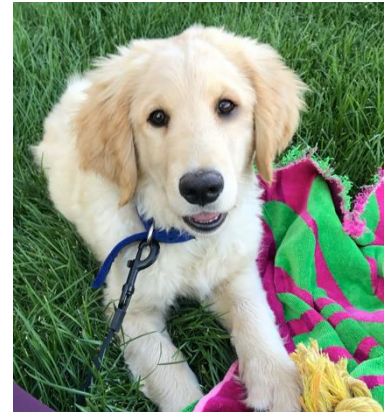
Earthworm Lengths in Millimeters (mm)									
6.0	11.2	18.3	19.4	20.2	23.1	23.3	25.2	25.7	26.0
27.8	27.9	28.4	29.9	32.5	33.2	41.1	42.3	48.5	52.1
54.1	59.7	60.3	77.6	93.2					

- Create an appropriate graphical representation for the data.
- Explain your reasoning for choosing your display type.
- If you were asked to create an additional display, which would you make and why?

13.8.4 Your Turn!

1. The birth weights, in ounces, of golden retriever puppies born at a kennel were recorded for the past month. The data is recorded in the table.

13	14	15	15	16	16	16	16	17
17	17	17	17	17	17	18	18	18
18	18	18	18	18	19	20		



- a. Create an appropriate graphical display for the data.
- b. Explain your reasoning for choosing that display type.
2. Tom Brady's pass-completion percentages from 2009 to 2020 are listed in the table.

Year	2009	2010	2011	2012	2013	2014
%	65.7	65.9	65.6	63	60.5	64.1
Year	2015	2016	2017	2018	2019	2020
%	64.4	67.4	66.3	65.8	60.8	65.7

- a. Create an appropriate graphical display for the data.

- b. Explain your reasoning for choosing that display type.

- c. Compare your graph with your partner. If you created the same type of graph, explain why this graph is the best choice to represent the data. If you created different displays, discuss which graph is a better choice.

13.8.5 Wrap-Up: Cool Down

1. What is your favorite type of graphical representation to create when displaying numerical data?
2. Write a note to your teacher explaining why this is your favorite type of display to create. Be specific in your explanation. You can use illustrations and examples from the lesson to help in your explanation.

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